Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**5**
Eigenstate idx	**2 (1-indexed: 3)**

Eigenstate properties

Energy E0	**-2.51637687**
Energy E1	**-2.42291126**
Gap (E1-E0)	**0.093466**
<S^2>	**-0.000000**
S	**-0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.442637**
<D> / site	**0.073773**

RDM and GME statistics for this run (5-body clusters)

Hilbert space dim : 2^5 = 32
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.421663 [min 0.421663 , max 0.421663]
Von Neumann entropy : mean = 1.041395 [min 1.041395 , max 1.041395]
Rank of rho : mean = 8.00 [min 8 , max 8]
C_GME (best estimate) : mean = 0.733508 [min 0.730928 , max 0.734896]

NOTE Analysis performed on eigenstate index 2 (state 3 in 1-indexed convention), not the ground state.

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Summary: All 5-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-2.516377

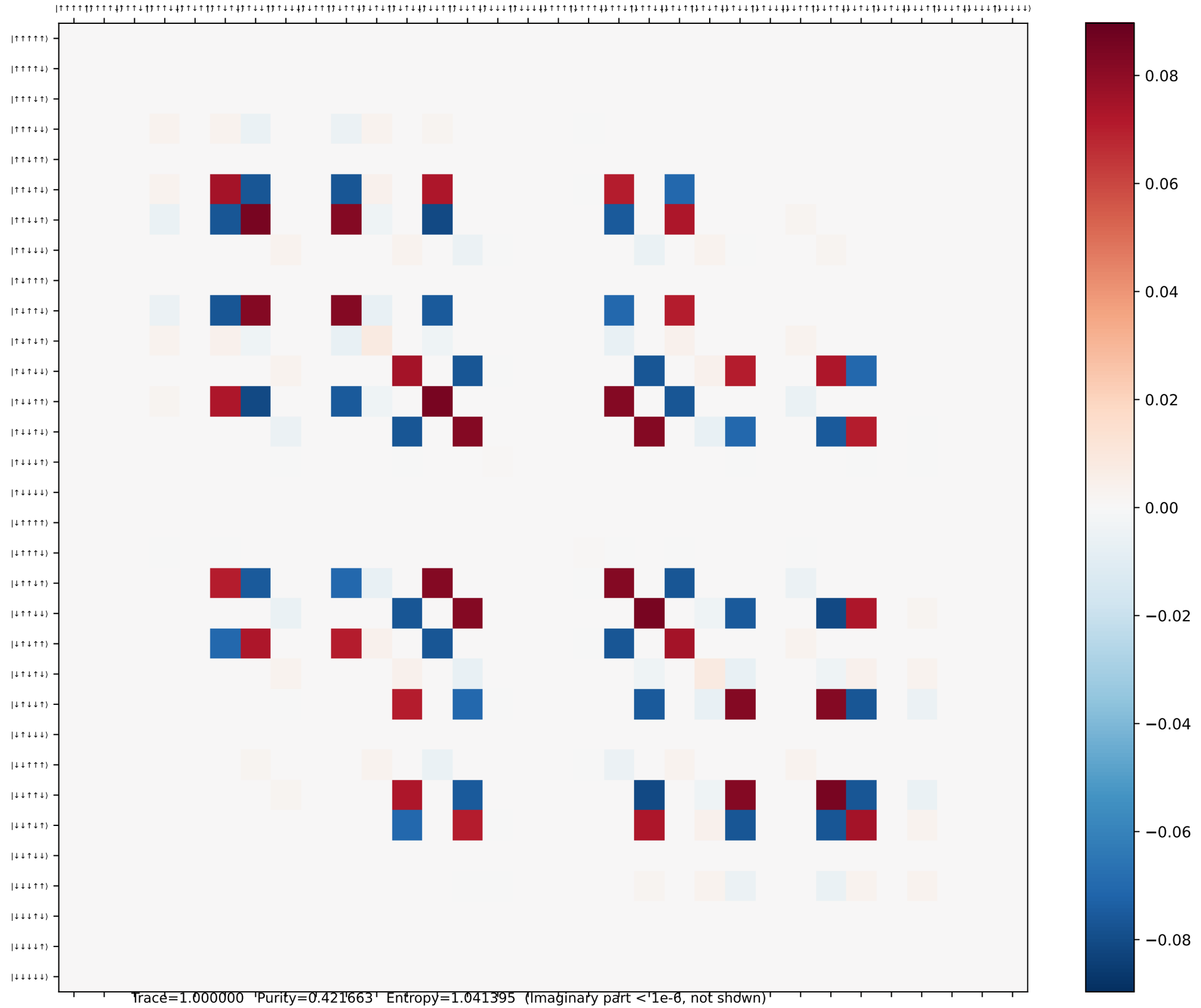
Sites	Purity	C_GME	C_LB	C_cayl	C_gcg	C_stiefel	C_UB	Status
(0, 1, 2, 3, 4)	0.421663	–	inf	0.000000	0.729798	0.734896	0.930914	MIXED
(1, 2, 3, 4, 5)	0.421663	–	inf	0.000000	0.729798	0.730928	0.930914	MIXED
(2, 3, 4, 5, 0)	0.421663	–	inf	0.000000	0.729798	0.733165	0.930914	MIXED
(3, 4, 5, 0, 1)	0.421663	–	inf	0.000000	0.729798	0.734062	0.930914	MIXED
(4, 5, 0, 1, 2)	0.421663	–	inf	0.000000	0.729798	0.733974	0.930914	MIXED
(5, 0, 1, 2, 3)	0.421663	–	inf	0.000000	0.729798	0.734024	0.930914	MIXED

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_gcg= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

5-Body RDM (Sites 0,1,2,3,4)

Real Part



Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -5.289080e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

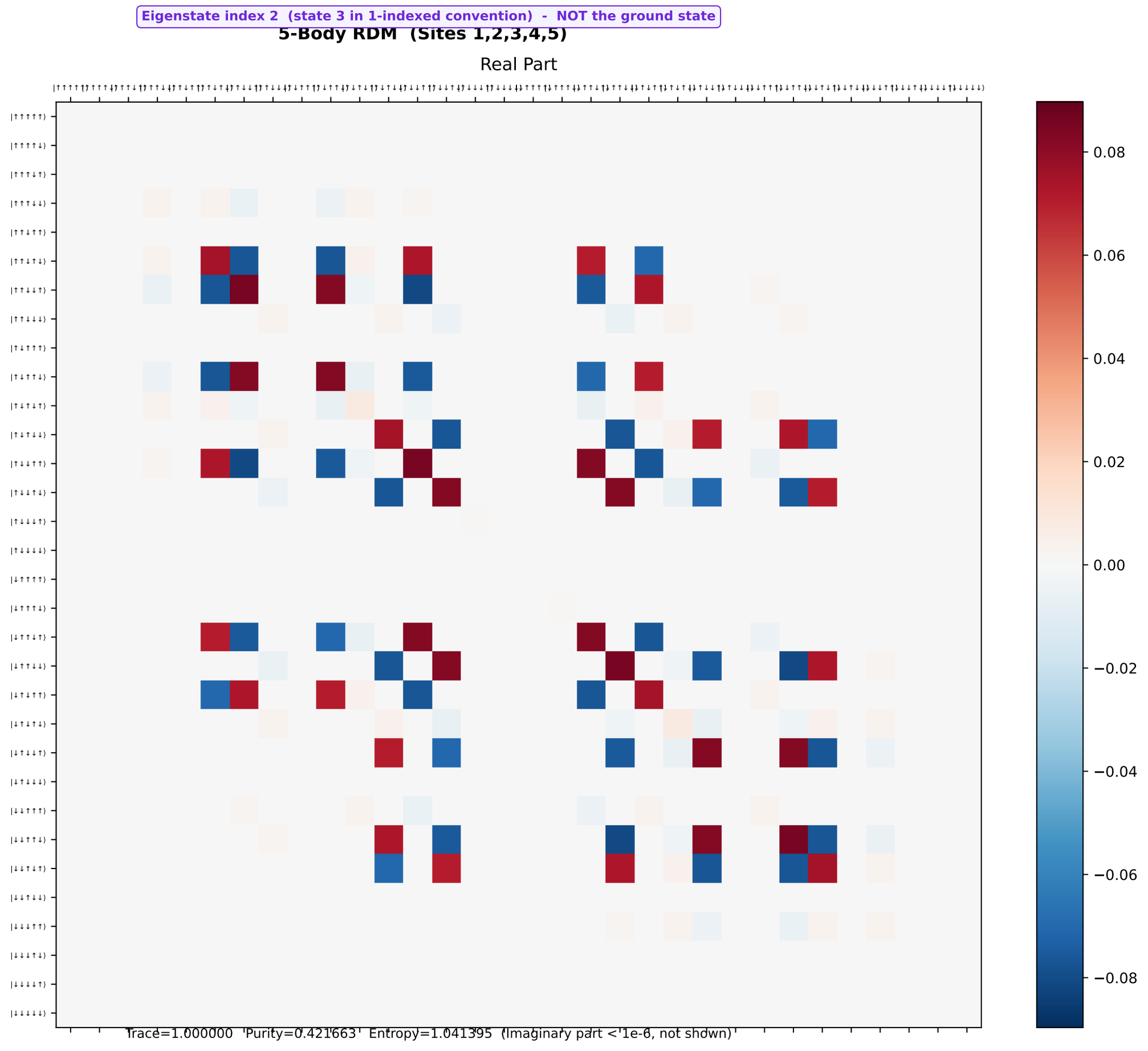
STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73489638   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```


Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state
5-Body RDM (Sites 1,2,3,4,5)
Real Part
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -5.214371e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

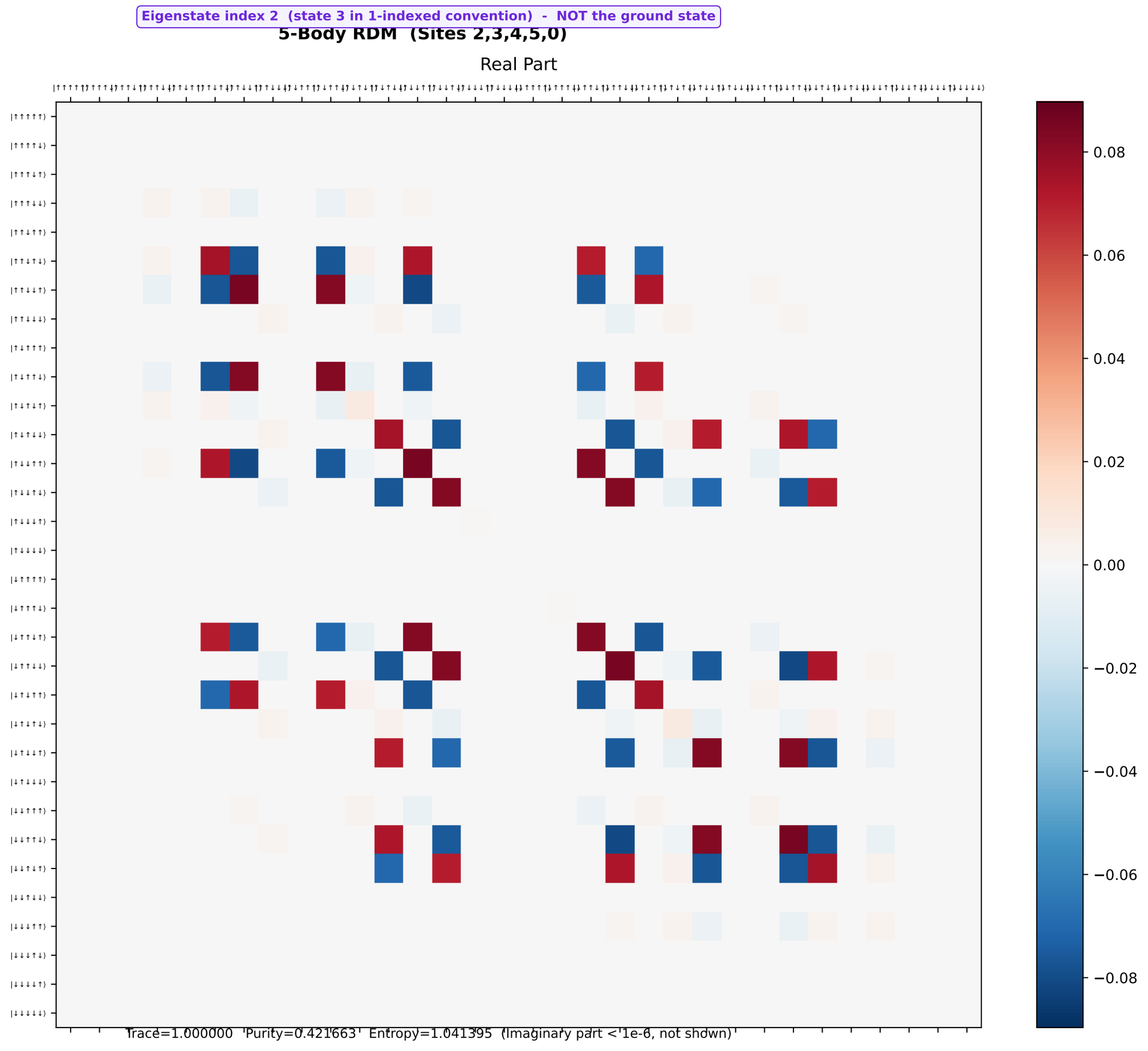
STATUS: MIXED STATE
___________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73092774   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```


Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state
5-Body RDM (Sites 2,3,4,5,0)
Real Part
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 2,3,4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -3.086959e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

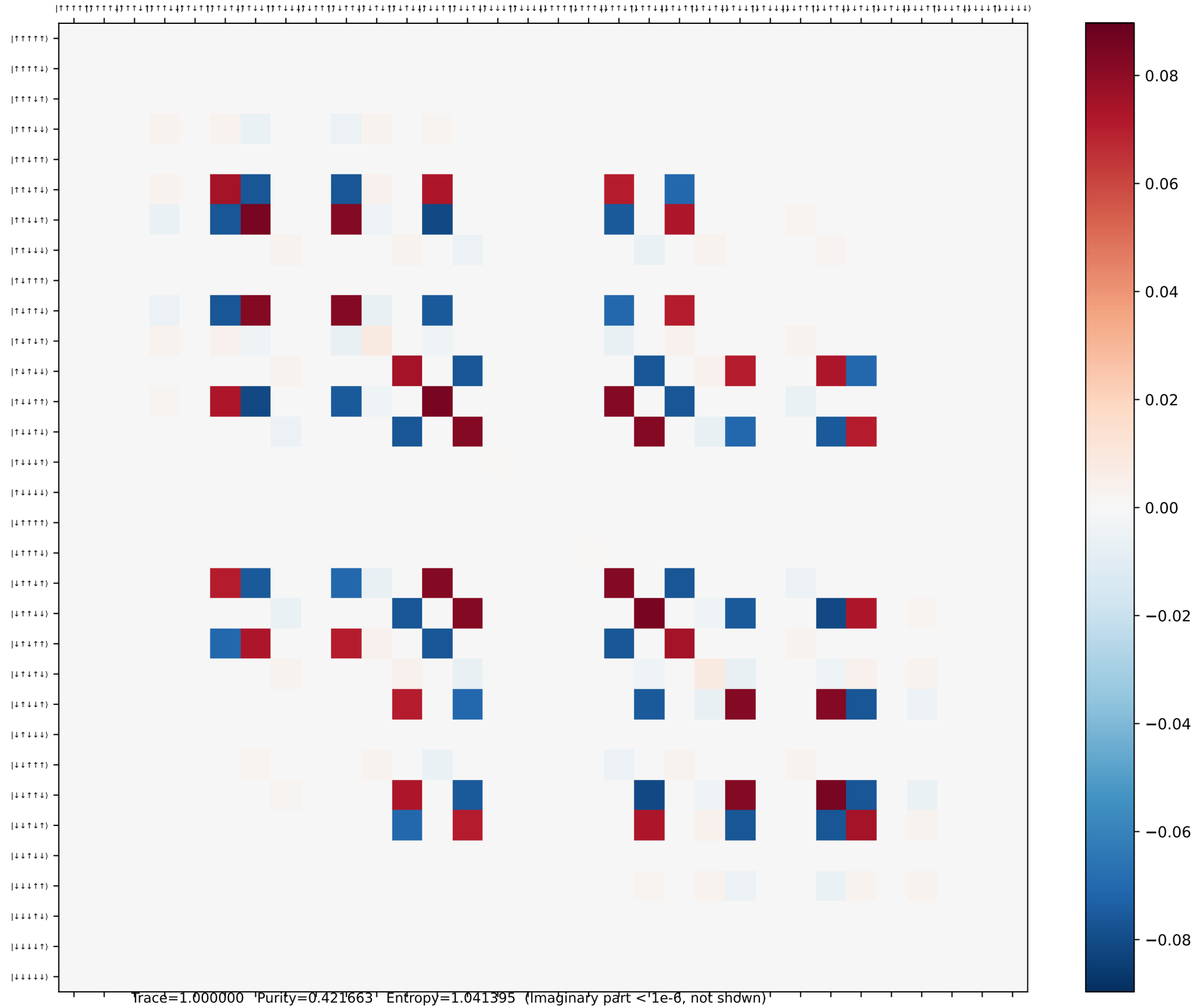
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73316529   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

5-Body RDM (Sites 3,4,5,0,1)

Real Part



Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 3,4,5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -2.921125e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

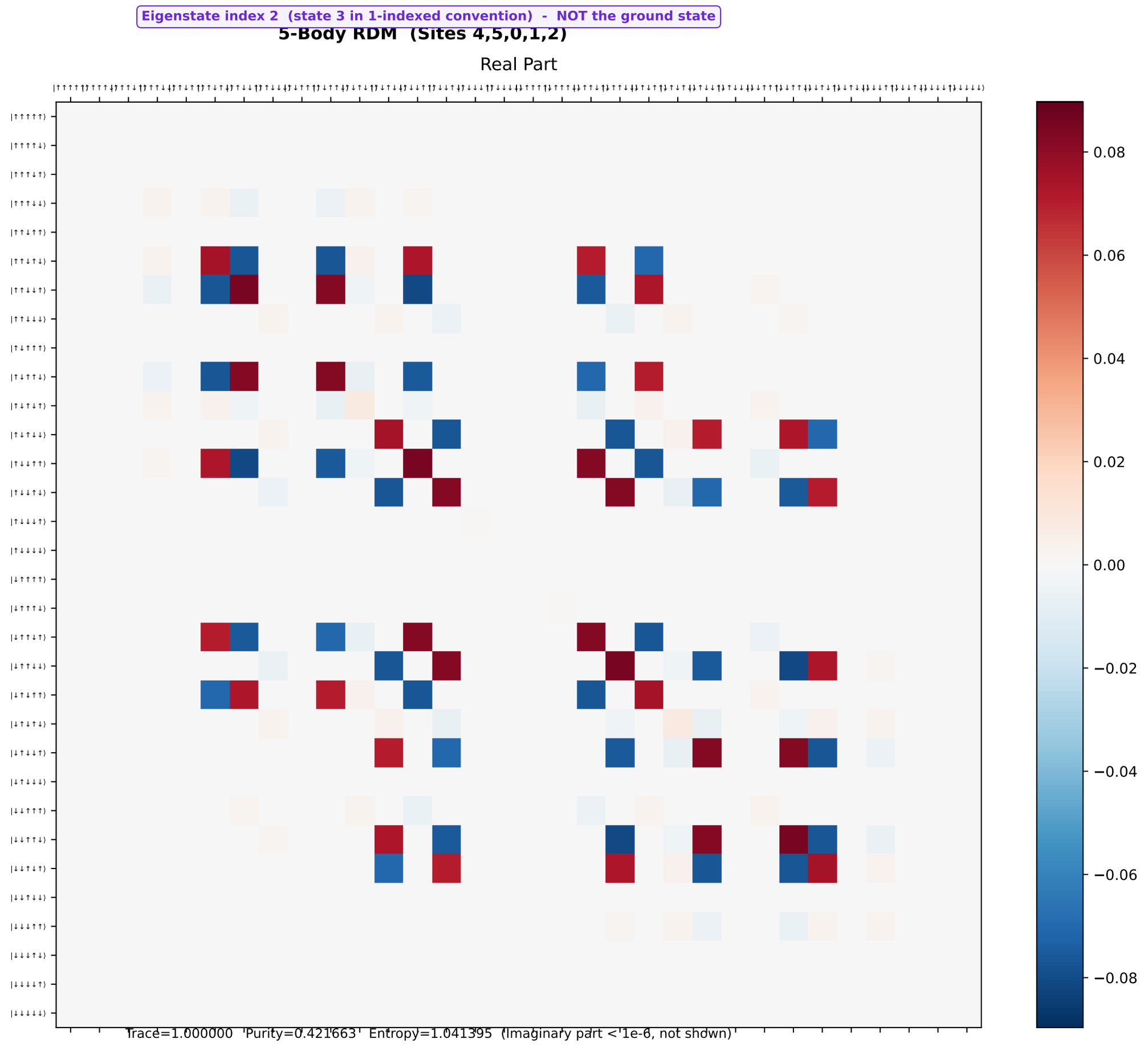
STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73406200   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```


Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state
5-Body RDM (Sites 4,5,0,1,2)
Real Part
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 4,5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -8.421314e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
___________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

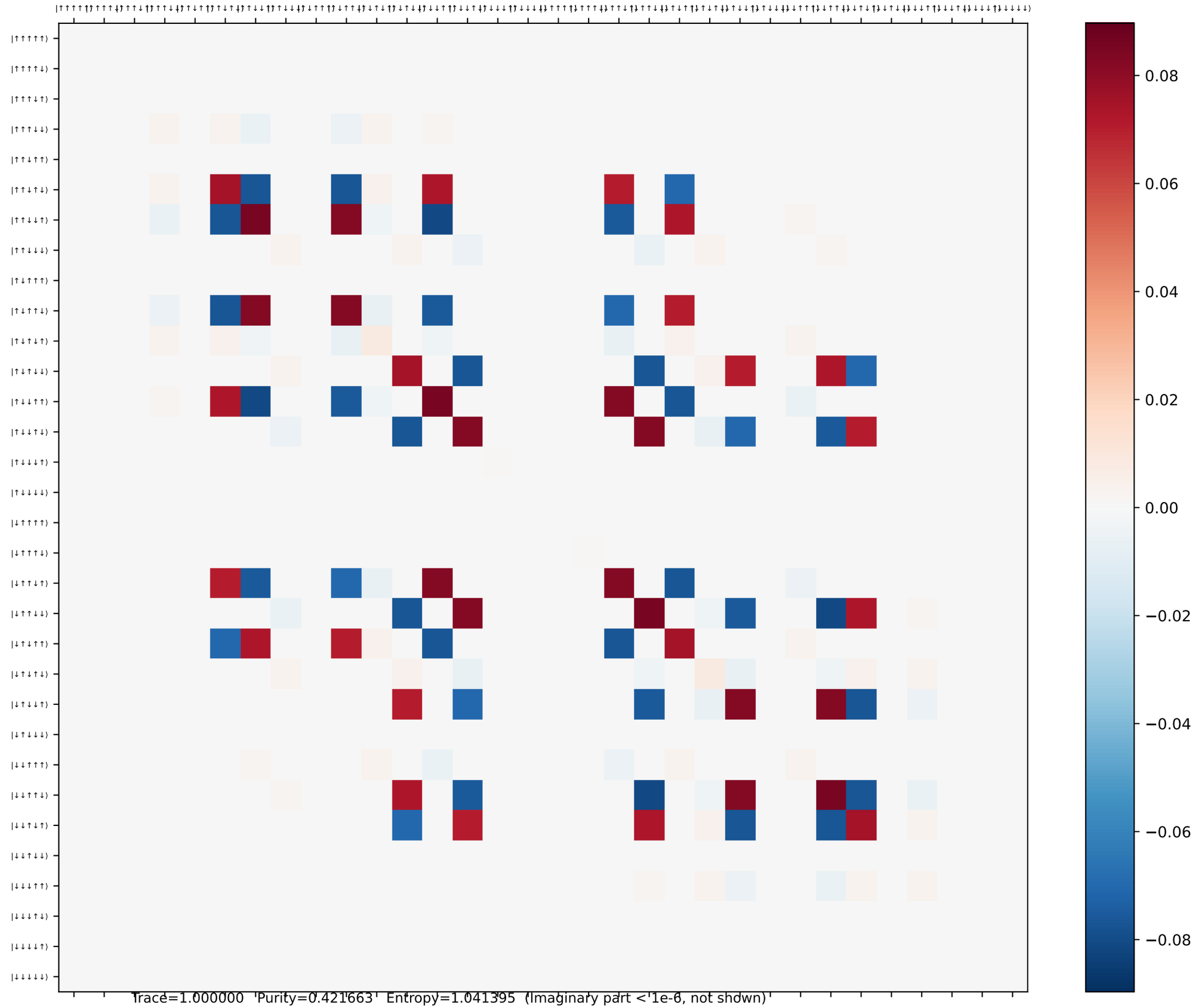
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73397445   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

5-Body RDM (Sites 5,0,1,2,3)

Real Part



Trace=1.000000 Purity=0.421663 Entropy=1.041395 (Imaginary part <1e-6, not shown)

Eigenstate index 2 (state 3 in 1-indexed convention) - NOT the ground state

Analysis Summary - 5-Body RDM (Sites 5,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4216629526
  Von Neumann entropy  : 1.0413953668
  Eigenvalue min       : -5.638452e-17
  Eigenvalue max       : 4.580292e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.42166295

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.73402415   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.93091433

  Status: nONE
```